ARTHROSCOPY : THE JOURNAL OF ARTHROSCOPIC & RELATED SURGERY : OFFICIAL PUBLICATION OF THE ARTHROSCOPY ASSOCIATION OF NORTH AMERICA AND THE INTERNATIONAL ARTHROSCOPY ASSOCIATION • 2026.0

Top 100 Orthopedic AI • Rank #4

Large Language Model Predicts Surgeon Recommendations for Imaging and Surgery for Patients Presenting for Knee and Shoulder Complaints With 70% and 81% Accuracy Using Previsit Questionnaire Responses.

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Journal:	Publication Year:	Citations:	DOI:
Arthroscopy : the journal of arthroscopic & related surgery : official publication of the Arthroscopy Association of North America and the International Arthroscopy Association	2026.0	1	10.1002/arj.70016

CLINICAL & AGENTIC AI TAKEAWAY

The pretrained large language model accurately predicts orthopaedic surgeon recommendations for imaging and surgery using previsit questionnaire responses for knee and shoulder complaints with 70% accuracy and 81% accuracy when augmented with radiology reports.

ABSTRACT

Purpose: To validate the performance of a pretrained large language model (LLM) in predicting orthopaedic surgeon recommendations for management of newly referred patients, using free-text previsit questionnaire responses as input. Methods: This retrospective cross-sectional study included new patients visiting an orthopaedic sports medicine clinic between 2020 and 2023. Using zero-shot prompting, the LLM analyzed previsit questionnaire responses (e.g., "When did you start to have pain?") to predict whether patients required advanced imaging and/or surgical intervention. The LLM was blinded to all other clinical information, including surgeon notes, physical exams, or referral data. Model predictions were evaluated with accuracy, sensitivity, and specificity in comparison to actual surgeon-generated plans. For a subset of patients who had undergone advanced imaging, the LLM was augmented with free-text radiology reports and asked to provide updated surgical recommendations. Results: In the combined cohort of 1141 patients, the LLM predicted surgeon recommendation for advanced imaging with 70% accuracy, 83% sensitivity, and 64% specificity using previsit questionnaire responses alone. Imaging predictions were accurate for common diagnoses, including anterior cruciate ligament (ACL, 94%), meniscus (85%), and rotator cuff (80%) injuries but poor for knee (54%) and shoulder arthritis (66%). When augmented with imaging reports, the LLM predicted recommendations for surgery with 81% accuracy, 88% sensitivity, and 72% specificity. Surgical predictions were highly accurate for ACL (93%), meniscus (78%), rotator cuff

(83%), and shoulder instability related pathologies (78%). Conclusions: Using previsit questionnaire data from new orthopaedic patients with knee and shoulder complaints, the pretrained LLM showed 70% accuracy for imaging recommendations, and the augmented surgical-decision LLM showed 81% accuracy for surgical recommendations. Level of evidence: Level III, retrospective diagnostic case-control study. © 2026 the Arthroscopy Association of North America.

Study Overview & Background

Purpose: To validate the performance of a pretrained large language model (LLM) in predicting orthopaedic surgeon recommendations for management of newly referred patients, using free-text previsit questionnaire responses as input. Methods: This retrospective cross-sectional study included new patients visiting an orthopaedic sports medicine clinic between 2020 and 2023. Using zero-shot prompting, the LLM analyzed previsit questionnaire responses (e.g., "When did you start to have pain?") to predict whether patients required advanced imaging and/or surgical intervention. The LLM was blinded to all other clinical information, including surgeon notes, physical exams, or referral data. Model predictions were evaluated with accuracy, sensitivity, and specificity in comparison to actual surgeon-generated plans. For a subset of patients who had undergone advanced imaging, the LLM was augmented with free-text radiology reports and asked to provide updated surgical recommendations. Results: In the combined cohort of 1141 patients, the LLM predicted surgeon recommendation for advanced imaging with 70% accuracy, 83% sensitivity, and 64% specificity using previsit questionnaire responses alone. Imaging predictions were accurate for common diagnoses, including anterior cruciate ligament (ACL, 94%), meniscus (85%), and rotator cuff (80%) injuries but poor for knee (54%) and shoulder arthritis (66%). When augmented with imaging reports, the LLM predicted recommendations for surgery with 81% accuracy, 88% sensitivity, and 72% specificity. Surgical predictions were highly accurate for ACL (93%), meniscus (78%), rotator cuff (83%), and shoulder instability related pathologies (78%). Conclusions: Using previsit questionnaire data from new orthopaedic patients with knee and shoulder complaints, the pretrained LLM showed 70% accuracy for imaging recommendations, and the augmented surgical-decision LLM showed 81% accuracy for surgical recommendations. Level of evidence: Level III, retrospective diagnostic case-control study. © 2026 the Arthroscopy Association of North America.

Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

The pretrained large language model accurately predicts orthopaedic surgeon recommendations for imaging and surgery using previsit questionnaire responses for knee and shoulder complaints with 70% accuracy and 81% accuracy when augmented with radiology reports.

Extended Repository Abstract

Purpose To validate the performance of a pretrained large language model (LLM) in predicting orthopaedic surgeon recommendations for management of newly referred patients, using free-text previsit questionnaire responses as input. Methods This retrospective cross-sectional study included new patients visiting an orthopaedic sports medicine clinic between 2020 and 2023. Using zero-shot prompting, the LLM analyzed previsit questionnaire responses (e.g., "When did you start to have pain?") to predict whether patients required advanced imaging and/or surgical intervention. The LLM was blinded to all other clinical information, including surgeon notes, physical exams, or referral data. Model predictions were evaluated with accuracy, sensitivity, and specificity in comparison to actual surgeon-generated plans. For a subset of

patients who had undergone advanced imaging, the LLM was augmented with free-text radiology reports and asked to provide updated surgical recommendations. Results In the combined cohort of 1141 patients, the LLM predicted surgeon recommendation for advanced imaging with 70% accuracy, 83% sensitivity, and 64% specificity using previsit questionnaire responses alone. Imaging predictions were accurate for common diagnoses, including anterior cruciate ligament (ACL, 94%), meniscus (85%), and rotator cuff (80%) injuries but poor for knee (54%) and shoulder arthritis (66%). When augmented with imaging reports, the LLM predicted recommendations for surgery with 81% accuracy, 88% sensitivity, and 72% specificity. Surgical predictions were highly accurate for ACL (93%), meniscus (78%), rotator cuff (83%), and shoulder instability related pathologies (78%). Conclusions Using previsit questionnaire data from new orthopaedic patients with knee and shoulder complaints, the pretrained LLM showed 70% accuracy for imaging recommendations, and the augmented surgical-decision LLM showed 81% accuracy for surgical recommendations. Level of evidence Level III, retrospective diagnostic case-control study.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

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- **Journal / Venue:** Arthroscopy : the journal of arthroscopic & related surgery : official publication of the Arthroscopy Association of North America and the International Arthroscopy Association (2026.0)
- **SJR Metric:** Quartile 1.0
- **Citation Count:** 1 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.1002/arj.70016
- **Consensus Evidence Link:** https://consensus.app/papers/large-language-model-predicts-surgeon-recommendations-halvorson-keeley/6746a55136c3577fad1a705f2dd7215b/

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